Venturi Flow Calculation

Problem Statement

F1_7: A horizontal pipe with a diameter of 120 mm has a reduced section of 60 mm, forming a venturi. The pressure difference between the 120 mm and 60 mm sections is measured using a mercury-filled U-tube. When the pipe is filled with seawater flowing through it, the mercury level difference is 440 mm. Friction losses are negligible.

A. The velocity of the water in the 120 mm section.

B. The mass flow rate in tonnes per hour.

Note:

- Relative Density of Mercury = 13.6
- Relative Density of Seawater = 1.025

Given Data

- **Diameter of the larger section:** $D_1 = 120$ mm $= 0.12$ m
- **Diameter of the smaller section:** $D_2 = 60$ mm $= 0.06$ m
- **Height difference of mercury column:** $h = 440$ mm $= 0.44$ m
- **Density of mercury:** $\rho_m = 13600$ kg/m³
- **Density of seawater:** $\rho = 1025$ kg/m³
- **Gravitational acceleration:** $g = 9.81$ m/s²

Apply Bernoulli's Equation

From Bernoulli's principle:

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

where v_1 and v_2 are the velocities in the 120 mm and 60 mm sections, respectively.

The pressure difference $P_1 - P_2$ is given by the hydrostatic relation:

$$P_1 - P_2 = \rho_m gh - \rho gh$$

Since $\rho_m \gg \rho$, we approximate:

$$P_1 - P_2 \approx \rho_m gh$$

Substituting values:

$$P_1 - P_2 = (13600 \times 9.81 \times 0.44)$$

$$P_1 - P_2 = \boxed{58673.28 \text{ Pa}}\ 2\ marks$$

Apply the Continuity Equation

$$A_1 v_1 = A_2 v_2$$

where $A_1 = \frac{\pi}{4}D_1^2$ and $A_2 = \frac{\pi}{4}D_2^2$.

$$A_1 = \frac{\pi}{4}(0.12)^2 = 0.0113 \text{ m}^2$$

$$A_2 = \frac{\pi}{4}(0.06)^2 = 0.00283 \text{ m}^2$$

Thus,

$$v_2 = \frac{A_1}{A_2}v_1 = \frac{0.0113}{0.00283}v_1 = \boxed{4v_1}\ 2\ marks$$

Solve for v_1

From Bernoulli's equation:

$$P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$$

$$58673.28 = \frac{1}{2} \times 1025 \times (16v_1^2 - v_1^2)$$

$$58673.28 = \frac{1}{2} \times 1025 \times 15v_1^2$$

$$58673.28 = 7678.125v_1^2$$

$$v_1 = \sqrt{\frac{58673.28}{7678.125}} = \boxed{2.77 \text{ m/s}} \; 2 \; marks$$

Calculate the Mass Flow Rate

$$\dot{m} = \rho Q = \rho A_1 v_1$$

$$\dot{m} = 1025 \times (0.0113 \times 2.77)$$

$$\dot{m} = \boxed{32.07 \text{ kg/s}} \; 2 \; marks$$

Convert to **tonnes per hour**:

$$\dot{m} = 32.07 \times 3600/1000$$

$$\dot{m} = \boxed{115.45 \text{ tonnes/hour}} \; 2 \; marks$$

Final Answers

A. **Velocity in the 120 mm section:** $v_1 = 2.77$ m/s

B. **Mass flow rate:** 115.45 tonnes/hour